MATH 1342 – Elementary Statistics – Week 9: P-values and Statistical Significance

Oct 20 – 22 | 4.2 Measuring Evidence with p-values • 4.3 Determining Statistical Significance | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Describe how a randomization distribution is generated assuming H_0 is true, and identify its center as the null value. (CLO 5, CLO 8) **2.** Compare sampling, bootstrap, and randomization distributions by their center and purpose. (CLO 5) **3.** Compute a p-value from a randomization distribution as a proportion of simulated statistics, choosing the correct tail(s) from H_a. (CLO 3, CLO 8) **4.** Interpret a p-value in context and relate its size to the strength of evidence against H_0. (CLO 4, CLO 8) **5.** Compare the p-value to α, decide whether to reject H_0, write a two-step conclusion in context, and avoid saying "accept H_0". (CLO 8)	e-Text: 4.2 Measuring Evidence with P-Values Interactive Lecture: 4.2 Measuring Evidence with P-Values Slides: 4.2 Measuring Evidence with P-Values e-Text: 4.3 Determining Statistical Significance Interactive Lecture: 4.3 Determining Statistical Significance Slides: 4.3 Determining Statistical Significance (includes "What is a p-value?" video) StatKey – Randomization Hypothesis Tests	Participation: Randomization Distributions – StatKey: dolphin study (16 flips) and tea vs. coffee re-randomization; count how many simulated statistics are as extreme as observed Participation: P-values and Conclusions – resveratrol examples (p = 0.04, 0.12, 0.0003): strength of evidence, decision at α, conclusion in context	Online Homework: 4.2 Measuring Evidence with P-Values (due Sun Oct 25, 11:59 PM) Online Homework: 4.3 Determining Statistical Significance (due Sun Oct 25, 11:59 PM) Take-Home Quiz 5 (due Tue Oct 27, 11:59 PM) – covers 3.4, 4.1–4.3	WileyPLUS StatKey Blackboard Zoom (live class) Assignment Tool in Blackboard

CLO 3 Compute and interpret empirical and theoretical probabilities using the rules of probabilities and combinatorics. **CLO 4** Explain the role of probability in statistics. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 8** Perform hypothesis testing using statistical methods.